DSA Experiment 3

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INPUT :-

```
#include <stdio.h>

void bubblesort(int arr[], int n);
void insertionsort(int arr[], int n);
void selectionsort(int arr[], int n);

void bubblesort(int arr[], int n)
{
    int i, j, temp;

    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

void insertionsort(int arr[], int n)
{
    int i, j, temp;

    for (i = 1; i <= n - 1; i++)
    {
        temp = arr[i];
```

```
        j = i - 1;

        while (j >= 0 && arr[j] > temp)
        {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = temp;
    }
}

void selectionsort(int arr[], int n)
{
    int i, j, temp, min;

    for (i = 0; i < n - 1; i++)
    {
        min = i;

        for (j = i + 1; j < n; j++)
        {
            if (arr[j] < arr[min])
            {
                min = j;
            }
        }

        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
    }
}

int main()
{
    int arr[100], n, i, choice;

    printf("Enter the number of elements: ");
    scanf("%d", &n);
```

```
printf("Enter elements: ");
for (i = 0; i < n; i++)
{
   scanf("%d", &arr[i]);
}

printf("\n1. Bubble Sort");
printf("\n2. Insertion Sort");
printf("\n3. Selection Sort");

printf("\nEnter your choice: ");
scanf("%d", &choice);

switch (choice)
{
   case 1:
      bubblesort(arr, n);
      break;

   case 2:
      insertionsort(arr, n);
      break;

   case 3:
      selectionsort(arr, n);
      break;

   default:
      printf("Invalid Choice");
      return 0;
}

printf("Sorted array: ");

for (i = 0; i < n; i++)
{
   printf("%d ", arr[i]);
}
```

```
    return 0;
}
```

OUTPUT:-

```
Enter the number of elements: 6
Enter elements: 1
2
3
7
8
5

1. Bubble Sort
2. Insertion Sort
3. Selection Sort
Enter your choice: 1
Sorted array: 1 2 3 5 7 8

=== Code Execution Successful ===
```